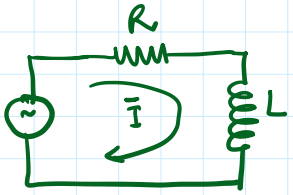
Lecture 17: Single Phase AC Circuits (Contd...)

RL-Series Circuit



$v(t) = V_m \sin \omega t$

$Z = R + j\omega L \; ; \; \bar{I} = \frac{\bar{V}}{Z} =$

$\bar{I} = \frac{V \angle 0}{(R + j\omega L)}$

$= \frac{V \angle 0}{\sqrt{R^2 + (\omega L)^2} \angle \tan^{-1} \frac{\omega L}{R}}$

$\bar{I} = \frac{V}{\sqrt{R^2 + (\omega L)^2}} \angle -\tan^{-1}\left(\frac{\omega L}{R}\right)$

$\downarrow$

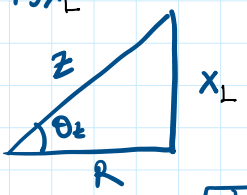
$i(t) = \frac{\sqrt{2}V}{\sqrt{R^2 + (\omega L)^2}} \sin\left[\omega t - \tan^{-1}\left(\frac{\omega L}{R}\right)\right]$

$i(t) = I_m \sin(\omega t - \phi)$

$Z = |Z| \angle \theta_z$

$= |Z| \cos\theta_z + |Z| \sin\theta_z$

$= R + jX_L$

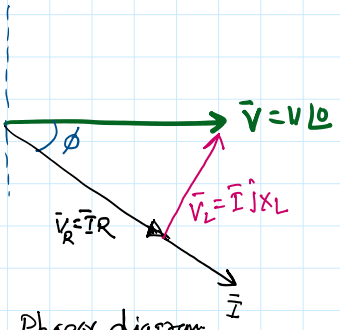


$Z = R + jX = \sqrt{R^2 + X_L^2} \angle \tan^{-1}\left(\frac{X_L}{R}\right)$

$\bar{V} = \bar{V}_R + \bar{V}_L$

$= \bar{I}R + \bar{I}(j\omega L)$

$= \bar{I}(R + j\omega L) = \bar{I}Z$



Phasor diagram

$\bar{S} = \bar{V}\bar{I}^* = P + jQ$

$= V\angle 0 * \frac{V}{\sqrt{R^2 + (\omega L)^2}} \left(\angle -\tan^{-1}\left(\frac{\omega L}{R}\right)\right)^*$

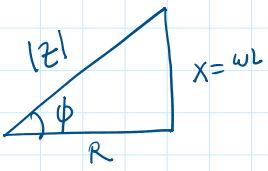
$= \frac{V^2}{|Z|} \angle \phi \qquad \phi = \tan^{-1}\left(\frac{\omega L}{R}\right)$

$= \frac{V^2}{|Z|}\cos\phi + j\frac{V^2}{|Z|}\sin\phi$

$= \frac{V^2}{|Z|^2} * R + j\frac{V^2}{|Z|} \cdot \frac{X_L}{|Z|}$

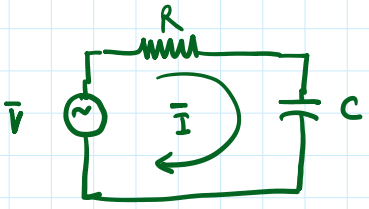
$= (I^2R) + j(I^2X_L) \Rightarrow P = VI\cos\phi = I^2R$

$Q = VI\sin\phi = I^2X_L$



$pf = \cos\phi = \frac{R}{\sqrt{R^2 + (\omega L)^2}}$

RC Series Circuit



$\bar{I} = \frac{\bar{V}}{Z} = \frac{\bar{V}}{R - \frac{j}{\omega C}}$

$V\angle 0 \quad \angle \tan^{-1}\left(\frac{1}{\omega RC}\right)$

$$\bar{I} = \frac{\bar{V}}{Z} \qquad Z = R - \frac{j}{\omega C}$$

$$= \frac{V\angle 0}{\sqrt{R^2+\left(\frac{1}{\omega C}\right)^2}\ \angle \tan^{-1}\left(\frac{1}{\omega RC}\right)}$$

$$= \frac{V}{\sqrt{R^2+\left(\frac{1}{\omega C}\right)^2}} \cdot \angle -\phi$$

$$\begin{cases} Z_1 = R \\ Z_2 = -j/\omega C \end{cases}$$

$$Z = Z_1 + Z_L = R - \frac{j}{\omega C} = R - jX_C$$

$$\phi_z = -\tan^{-1}\frac{1}{\omega RC}$$

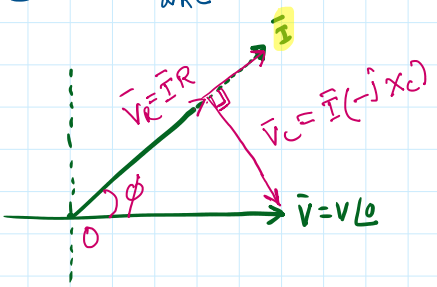
$$i(t) = \left(\frac{\sqrt{2}\,V}{|Z|}\right) \sin\left(\omega t + \tan^{-1}\frac{1}{\omega RC}\right)$$

$$i(t) = I_m \sin\left[\omega t + \tan^{-1}\left(\frac{1}{\omega RC}\right)\right]$$

$$\bar{V} = \bar{V}_R + \bar{V}_C$$

$$= \bar{I}\left(R - \frac{j}{\omega C}\right)$$

$$= \bar{I}R + \left[\bar{I}(-jX_C)\right]$$



Powers:

$$\bar{S} = \bar{V}\bar{I}^* = V\angle 0 \times \left(I\angle\phi\right)^*$$

$$= VI\angle -\phi = VI\cos\phi + j(-VI\sin\phi)$$

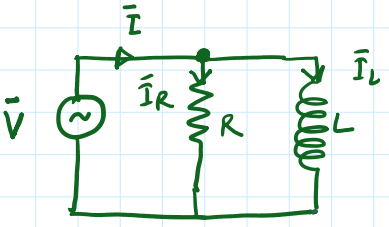
$$= VI \times \frac{R}{|Z|} + j\left(-VI \cdot \frac{X_C}{|Z|}\right)$$

$$= I^2R + j(-I^2X_C)$$

$$P = I^2R$$
$$Q = VI\sin\phi = -I^2X_C$$

RL Parallel Circuit:

KCL: $v(t) = \sqrt{2}V\sin\omega t$, $\bar{V} = V\angle 0$ V



$$i(t) = i_R(t) + i_L(t)$$

KCL: $\bar{I} = \bar{I}_R + \bar{I}_L$

$$= \frac{\bar{V}}{R} + \frac{\bar{V}}{j\omega L}$$

$$= \left(\frac{1}{R} + \frac{1}{j\omega L}\right)\bar{V} = \frac{1}{R} + j\left(-\frac{1}{\omega L}\right)$$

$$= (G + jB_L)\bar{V}$$

G = Conductance (resistance Ω^{-1}; mho)

B_L = Susceptance of Inductance $= -\frac{1}{\omega L}$

$$\bar{I} = Y\cdot\bar{V}$$

Y = Admittance $= \frac{1}{Z} = \frac{1}{\text{impedance}}$

$$\frac{1}{\quad} = \frac{1}{\quad} + \frac{1}{\quad}$$

KVL and KCL in AC Circuits

$$\sum_{j=1}^{K} \bar{V}_j = 0$$

$$\sum_{j=1}^{K} \bar{I}_j = 0$$

$Y = \text{Admittance} = \frac{1}{Z} = \frac{1}{\text{Impedance}}$

$Z = \frac{Z_1 Z_2}{Z_1 + Z_2}$

$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$

$Y = Y_1 + Y_2 = G + jB$

$Z = R + jX$; $Y = G + jB$

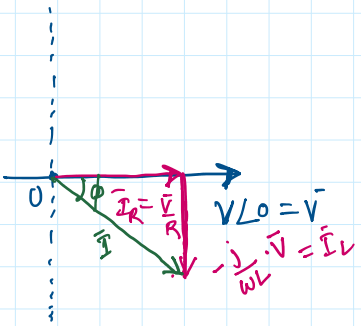
$Y = \frac{1}{Z} = \frac{1}{(R+jX)} \frac{(R-jX)}{(R-jX)}$

$= \frac{R-jX}{R^2-(jX)^2} = \frac{R-jX}{R^2+X^2}$

$= \frac{R}{R^2+X^2} + j\left(\frac{-X}{R^2+X^2}\right)$

$= G + jB$

$G = \frac{R}{R^2+X^2}$; $B = \frac{-X}{R^2+X^2}$



$P = ? \Rightarrow \bar{S} = \bar{V}\,\bar{I}^*$

$= \bar{V} * \left(\bar{V}(G+jB)\right)^*$

$= \underbrace{\bar{V} \cdot \bar{V}^*}\,(G+jB)^*$

$= V^2 (G - jB)$

$= V^2 G + j(-V^2 B_L)$

$= P + jQ \Rightarrow Q = -V^2 B_L$

$\bar{I} = \bar{V} Y$

$\bar{V}\bar{V}^* = V^2$

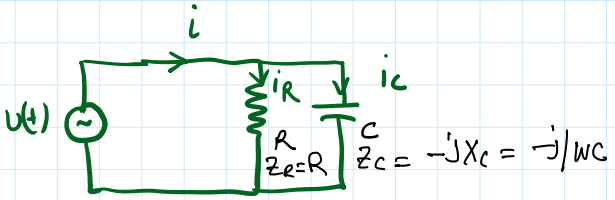
Remember B_L is negative for inductance:

$Z_L = j\omega L$

$Y_L = \frac{1}{Z_L} = -\frac{j}{\omega L}$

$= j\left(-\frac{1}{\omega L}\right) = jB_L$

RC Parallel Circuit



$U(t) = \sqrt{2}\, V \sin \omega t$

$\bar{V} = V\angle 0$

$\bar{I} = \bar{I}_R + \bar{I}_C$ → $i(t) = \sqrt{2}\, I \sin(\omega t + \angle \bar{I})$

$= \frac{\bar{V}}{R} + \frac{\bar{V}}{-j/\omega C}$

$= \left(\frac{1}{R} + j\omega C\right)\bar{V} = (G + jB_C)\,\bar{V}$

$G = \frac{1}{R}$; B_C = Susceptance of capacitance $= \omega C$

$\bar{S} = \bar{V}\,\bar{I}^* = \bar{V} \cdot \left(\bar{V}(G+jB_C)\right)^*$

$= V^2(G - jB_C) = V^2 G - jV^2 B_C$

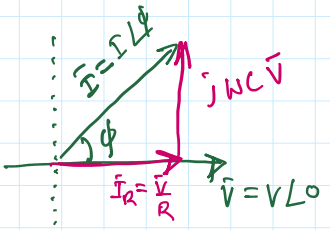
$= \frac{V^2}{R} + j(-V^2 B_C)$

$= P + jQ_C$

$P = \frac{V^2}{R} = V^2 G$; $Q_C = -V^2 B_C$

Q_C = Reactive power of capacitance is negative ⇒ it generates reactive power.

Q_C = Reactive power of capacitance is negative ⇒ it generates reactive power.



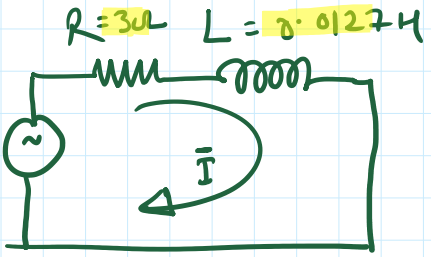
An Example:

$v(t) = 141 \sin 314t$ $= \sqrt{2}V \sin \omega t$ $\quad 2\pi \times 50 = 100\pi$

$\omega = 314t$

$R = 3\Omega$

$L = 0.0127H$



$\bar{I} = ?$ $\quad i(t) = ?$

P, Q, S, pf ?

$\bar{I} = \frac{\bar{V}}{\bar{z}}$, $z = R + j\omega L$; $X_L = 2\pi \times 50 \times 0.0127H$

$= 3 + j4$ $\quad = 4\Omega$

$z = 5\angle 53.13^\circ\ \Omega$

$\bar{I} = \frac{100\angle 0^\circ}{5\angle 53.13^\circ} = 20\angle -53.13^\circ$ A

$\downarrow$

$i(t) = 20\sqrt{2} \sin(\omega t - 53.13^\circ)$

$= I_m \sin(\omega t + \phi)$

$\bar{S} = \bar{V}\,\bar{I}^* = 100\angle 0 \times (20\angle -53.13^\circ)^*$

$= 2000\angle 53.13^\circ$

$= 2000\cos 53.13^\circ + j\,2000 \sin 53.13^\circ$

$= 1200 + j\,1600$

$= P + jQ$

$P = 1200$ Watts

$Q = 1600$ VArs

$S = 2000$ VA

$pf = \cos(-53.13^\circ) = 0.6$ (Lagging)

Phasor Diagram : $\bar{V}_R$, $\bar{V}_L$

$\bar{V}_R = \bar{I} \times R = 20\angle -53.13^\circ \times 3$

Phasor Diagram

$\bar{V}_R = \bar{I} * R = 20\angle -53.13^\circ * 3$

$= 60\angle -53.13^\circ$

$\bar{V}_L = \bar{I}(j\omega L)$

$= 20\angle -53.13^\circ \quad j4$

$= 80\angle 36.9^\circ \quad V$

